Find A Gene Project

Matthew Huang (PID: A17978767)

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1 Part 1:

1.1 Q1.

Tell me the name of a protein you are interested in. Include the species, accession number and known function. This can be a human protein or a protein from any other species as long as it's function is known.

Name: Kinesin Family Number 11 (KIF11)

Accession: NP_004514

Species: Homo Sapiens

Function: A motor protein required for the establishment of a bipolar mitotic spindle, thus facilitating chromosome congression during mitosis.

1.2 Q2.

Perform a BLAST search against a DNA database, such as a database consisting of genomic DNA or ESTs. The BLAST server can be at NCBI or elsewhere. Include details of the BLAST method used, database searched and any limits applied (e.g. Organism).

Method: tblastn search against ESTS

Database: Expressed Sequence Tags (est)

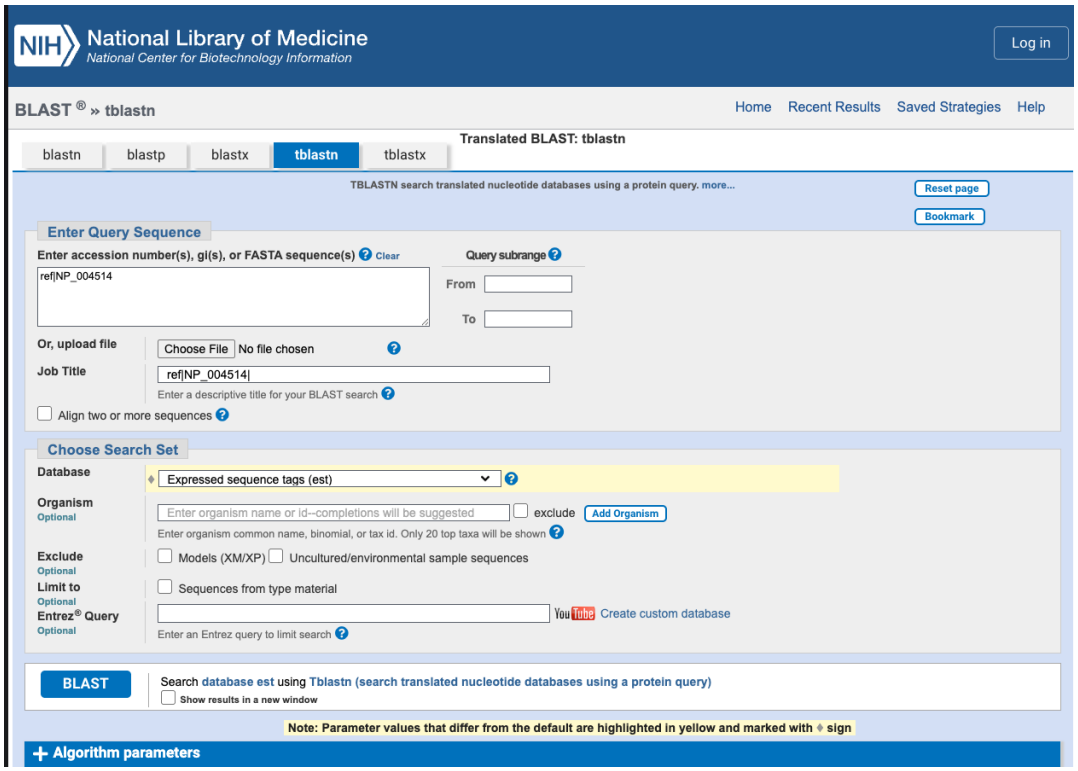


Figure 1: tBLASTn setup for human KIF11 (NP_004514) against the NCBI EST nucleotide database

Chosen Match: Accession FG708023.1, a 740 base pair expressed sequence tag (EST) from an Anolis carolinensis embryo cDNA library. Alignment details are shown below.

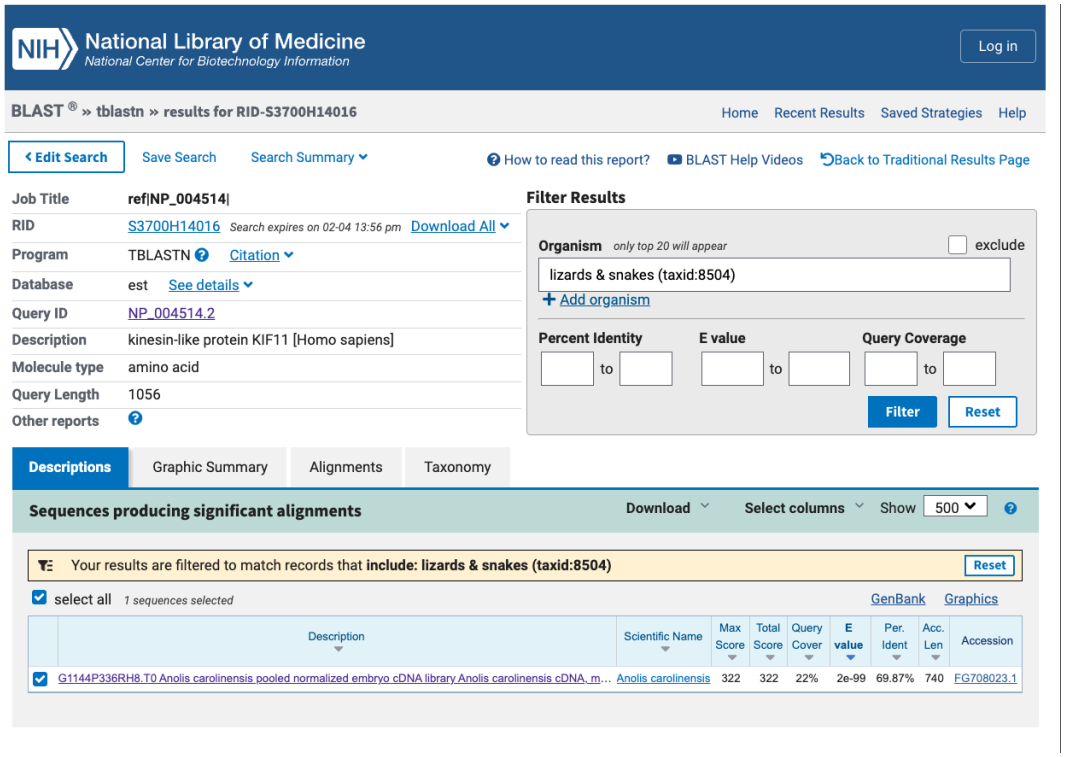


Figure 2: tBLASTn results summary showing a significant match between the human KIF11 protein (NP_004514) and an expressed sequence tag (FG708023.1) from *Anolis carolinensis* identified in the NCBI EST database.

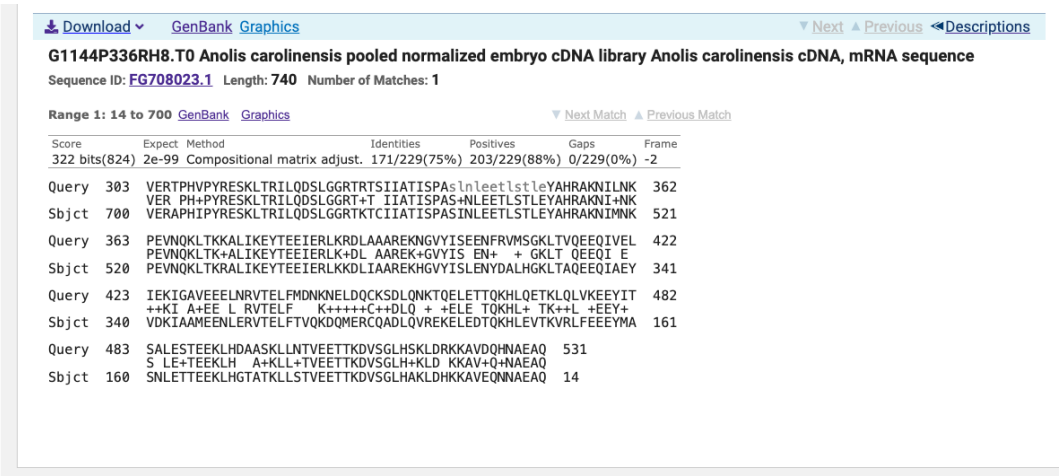


Figure 3: Pairwise tBLASTn alignment between the human KIF11 protein (NP_004514) and the *Anolis carolinensis* EST FG708023.1, showing alignment score, E-value, and sequence similarity.

1.3 Q3.

Gather information about this "novel" protein. At a minimum, show me the protein sequence of the "novel" protein as displayed in your BLAST results from [Q2] as FASTA format (you can copy and paste the aligned sequence subject lines from your BLAST result page if necessary) or translate your novel DNA sequence using a tool called EMBOSS Transeq at the EBI. Don't forget to translate all six reading frames; the ORF (open reading frame) is likely to be the longest sequence without a stop codon. It may not start with a methionine if you don't have the complete coding region. Make sure the sequence you provide includes a header/subject line and is in traditional FASTA format.

Chosen Sequence:

> *Anolis Carolinensis* VERAPHIPYRESKLTRILQDSLGGRTKTCIIATISPASIN-LEETLSTLEYAHRAKNIMNKPEVNQKLTKR ALIKEYTEEIERLKKDLI-AAREKHGVYISLENYDALHGKLTAQEEQIAEYVDKIAAMEENLERVTELFTV QKDQMERCQADLQVREKELEDTQKHLEVTKVRLFEEEYMASNLETTEEKL-HGTATKLLSTVEETTKDVSG LHAKLDHKKAVEQNNAEAQ

Name: Putatative KF11-like Protein

Species: *Anolis Carolinesis*

Taxonomy: Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Tetrapoda; Sauropsida; Lepidosauria; Squamata; Iguania; Dactyloidae; Anolis

1.4 Q4.

Prove that this gene, and its corresponding protein, are novel. For the purposes of this project, "novel" is defined as follows. Take the protein sequence (your answer to [Q3]), and use it as a query in a blastp search of the nr database at NCBI.

A tblastn search with a nr/nt database yielded a top hit result is to an Anolis Crolinesis protein with <100% percent identity. See attached screenshots for top hits and selected alignment details.

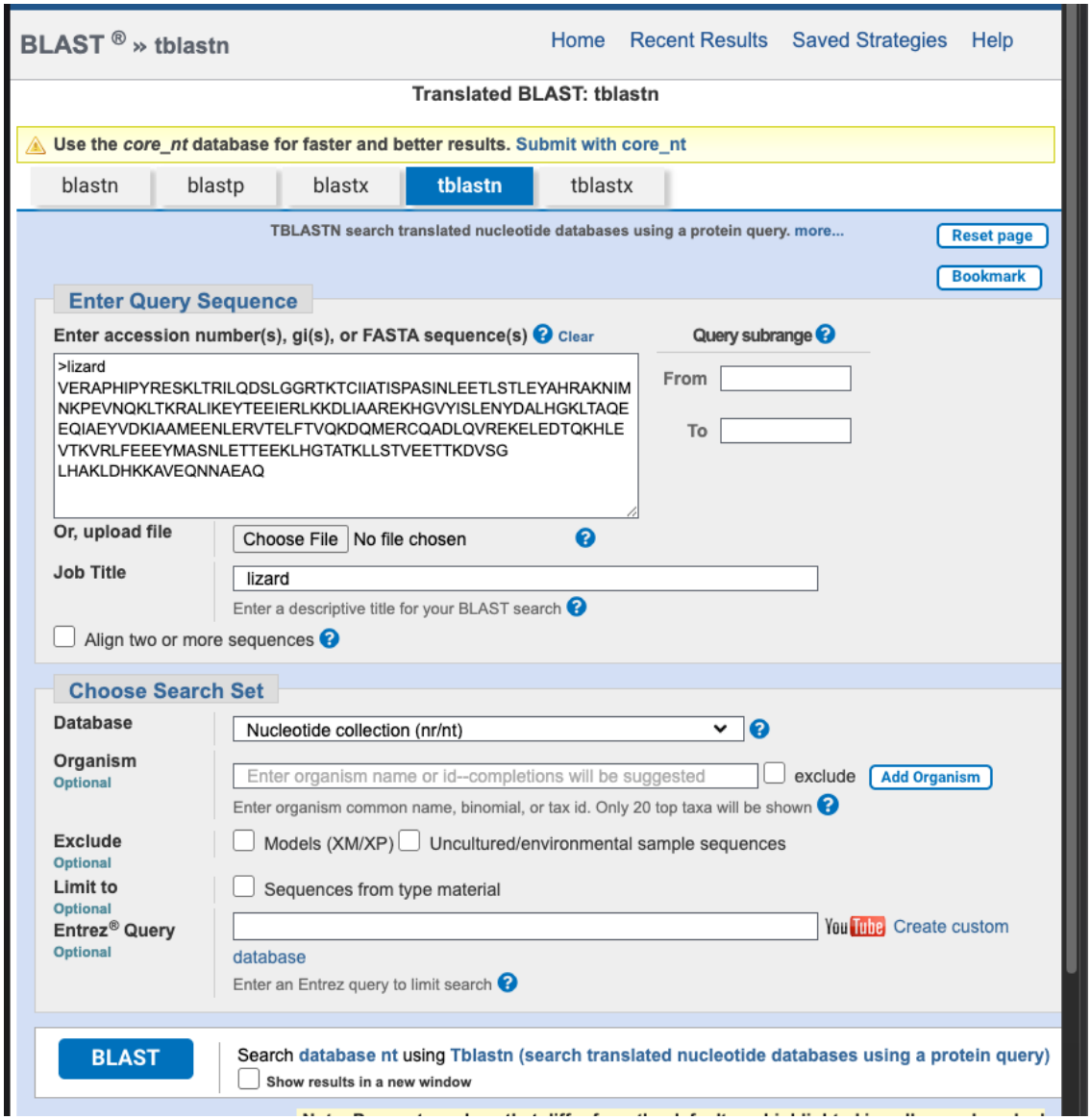
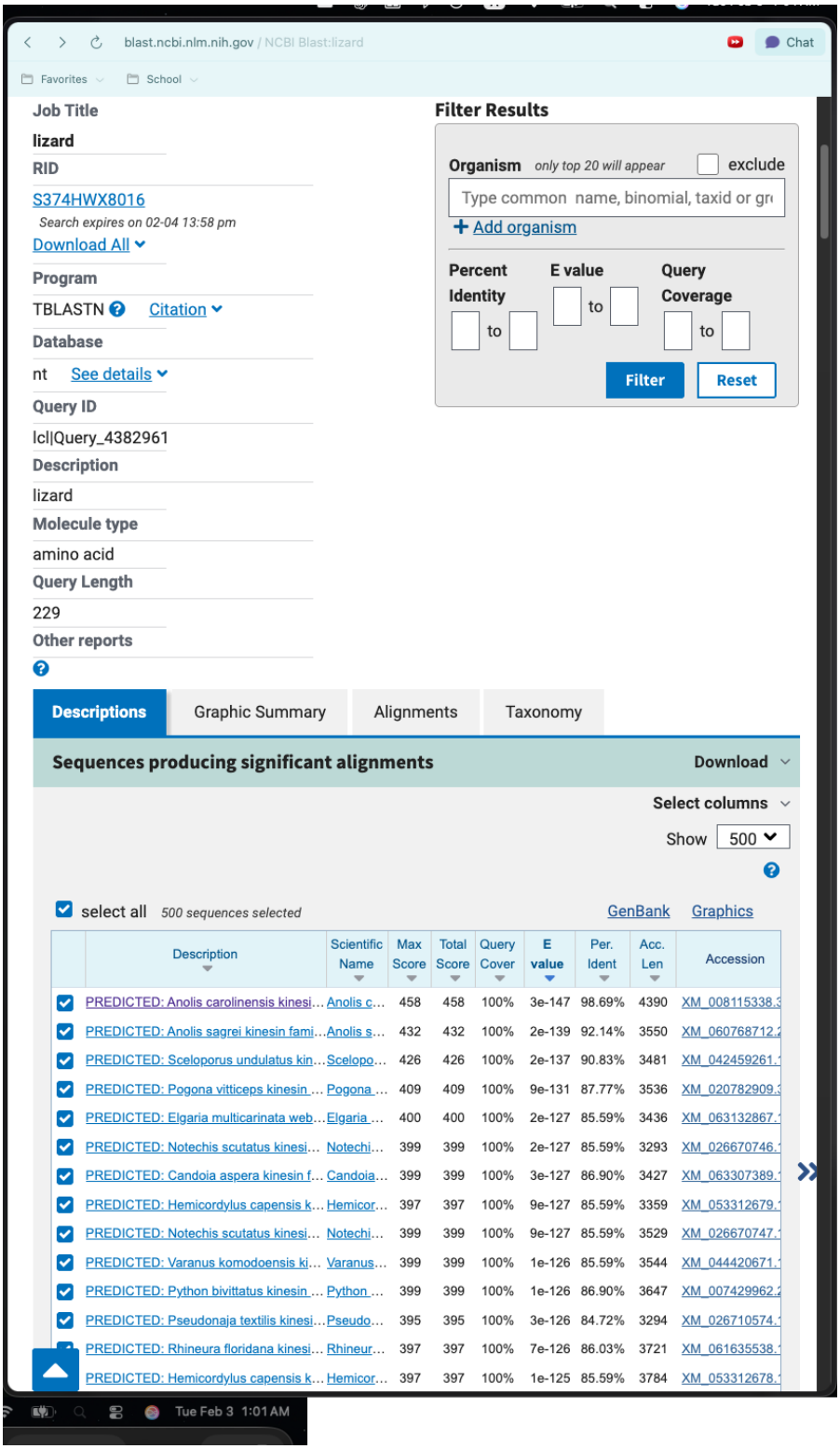


Figure 4: tBLASTn search setup using the translated 'lizard' protein query against the NCBI nucleotide collection (nt), used to search for matching coding regions in nucleotide sequences.



Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
PREDICTED: Anolis carolinensis kinesi...	Anolis c...	458	458	100%	3e-147	98.69%	4390	XM_008115338.3
PREDICTED: Anolis sagrei kinesin fami...	Anolis s...	432	432	100%	2e-139	92.14%	3550	XM_060768712.2
PREDICTED: Sceloporus undulatus kin...	Scelopo...	426	426	100%	2e-137	90.83%	3481	XM_042459261.1
PREDICTED: Pogona vitticeps kinesin ...	Pogona ...	409	409	100%	9e-131	87.77%	3536	XM_020782909.3
PREDICTED: Elgaria multicarinata web...	Elgaria ...	400	400	100%	2e-127	85.59%	3436	XM_063132867.1
PREDICTED: Notechis scutatus kinesi...	Notechi...	399	399	100%	2e-127	85.59%	3293	XM_026670746.1
PREDICTED: Candoia aspera kinesin f...	Candoia...	399	399	100%	3e-127	86.90%	3427	XM_063307389.1
PREDICTED: Hemicordylus capensis k...	Hemicor...	397	397	100%	9e-127	85.59%	3359	XM_053312679.1
PREDICTED: Notechis scutatus kinesi...	Notechi...	399	399	100%	9e-127	85.59%	3529	XM_026670747.1
PREDICTED: Varanus komodoensis ki...	Varanus...	399	399	100%	1e-126	85.59%	3544	XM_044420671.1
PREDICTED: Python bivittatus kinesin ...	Python ...	399	399	100%	1e-126	86.90%	3647	XM_007429962.2
PREDICTED: Pseudonaja textilis kinesi...	Pseudo...	395	395	100%	3e-126	84.72%	3294	XM_026710574.1
PREDICTED: Rhineura floridana kinesi...	Rhineur...	397	397	100%	7e-126	86.03%	3721	XM_061635538.1
PREDICTED: Hemicordylus capensis k...	Hemicor...	397	397	100%	1e-125	85.59%	3784	XM_053312678.1

Figure 5: BLAST results used to assess novelty of the putative protein sequence: top database hits show high similarities with full query coverage but less than 100% amino-acid identity.